

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lectures 9-10 (Ch. 5)

The IS-LM model

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies

Economic Bulletin No. 1 - 2025

**Euro-area GDP slows, inflation hovers around 2 per cent**

Based on the available information, economic growth in the euro area weakened in the fourth quarter of 2024, penalized by lacklustre consumption and investment and by declining exports. **The trend in manufacturing remains disappointing, especially in Germany;** the boost from services appears to have waned too. Inflation is still moderate, at around 2 per cent, with the core component being broadly stable, but it is still relatively high in services, partly reflecting delayed adjustments to past inflation. The Eurosystem staff projections for euro-area growth were revised downwards in December, to over 1 per cent per year on average for the three years 2025-27; inflation is set to stabilize around the European Central Bank's 2 per cent target.

The ECB cuts its key interest rates further

The **Governing Council of the ECB cut its reference rates by a further 25 basis points in December.** Markets expect an additional reduction of around 75 basis points over the course of 2025. Despite the gradual easing of monetary policy, credit growth in the euro area remains subdued, in an environment of high uncertainty and weak demand.

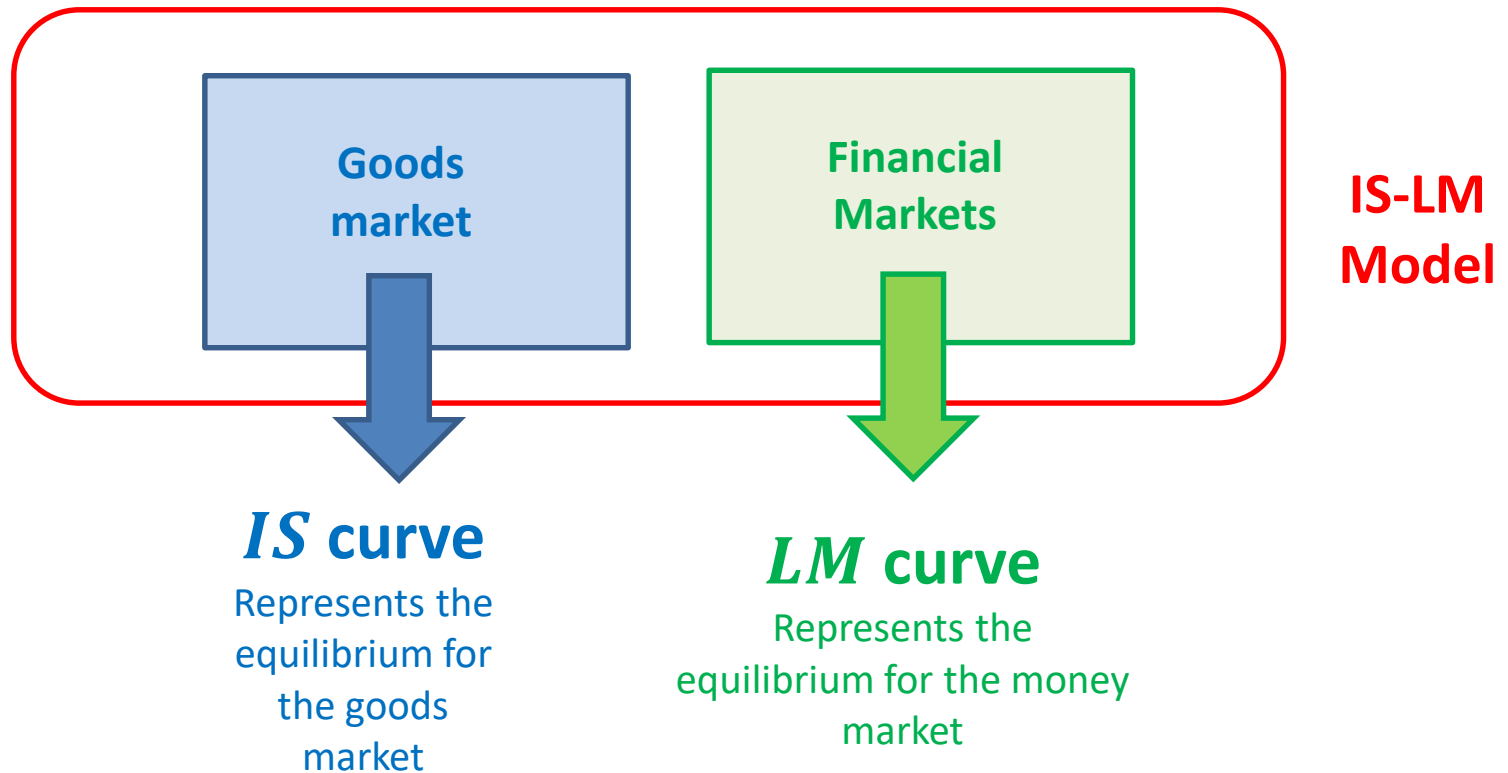
Growth is struggling to regain momentum in Italy

Economic activity in Italy remained weak in the fourth quarter of 2024, partly affected, as in the rest of the euro area, by the persistent sluggishness in manufacturing and the slowdown in services. In construction, the support provided by the works under the National Recovery and Resilience Plan was in contrast with the contraction in activity in the housing sector. Domestic demand was likely held back by the slowdown in household spending and by still unfavourable investment conditions. According to our projections, drawn up as part of the Eurosystem coordinated exercise, growth will gain momentum throughout the year, averaging around 1 per cent in the three years 2025-27.

January 2025

<https://www.bancaditalia.it/pubblicazioni/bollettino-economico/2025-1/index.html?com.dotmarketing.htmlpage.language=1>

We now have all the ingredients to build a model to explain the link btw real and financial markets in the short run when prices are fixed



Short-run Equilibrium:
the goods market and financial markets are simultaneously in equilibrium

MODEL IN A NUTSHELL

Main objectives:

1. Explain two key variables: GDP and interest rate in the short run in a closed economy
2. Understands the interdependence between the real and financial sides of the economy
3. Which policies should be implemented based on policymakers' objectives

THE LM CURVE

EQUILIBRIUM IN THE FINANCIAL MARKETS

Aggregate money supply must be equal to aggregate money demand

LM: EQUILIBRIUM ON FINANCIAL MARKETS

- The money market is in equilibrium when, given Y and P , the interest equates demand and supply for money

Nominal money demand $\leftarrow M^D = M^S \rightarrow$ **Nominal** money supply $\longrightarrow$ *Equilibrium in nominal terms*

$$\epsilon YL(i) = M$$

$$PYL(i) = M$$

Real money demand: D for money in terms of goods that can be bought

$YL(i) = \frac{M}{P}$ **Real** money supply $\longrightarrow$ *Equilibrium in real terms*

LM relation:

Pairs of Y and i such that financial markets are in equilibrium
 $\Rightarrow$ interest rate i that keeps the money market in equilibrium for any value of Y

What happens in the M market if Y changes?

It depends on the behavior of the CB wrt MP

Suppose that $Y \uparrow \Rightarrow$ more transactions $\Rightarrow M^d \uparrow \Rightarrow ED \Rightarrow$ agents sell bonds $\epsilon P_t \downarrow$ and $i \uparrow$

1. If the CB has a policy rate target, M^s must change in order to keep i constant: CB buys bonds, $\epsilon P_t \uparrow$ and $i \downarrow$, offsetting the behavior of the public. Endogenous MP (standard case)
2. If the CB does not have any policy rate target, M^s does not change and i increases. Exogenous MP (alternative case)

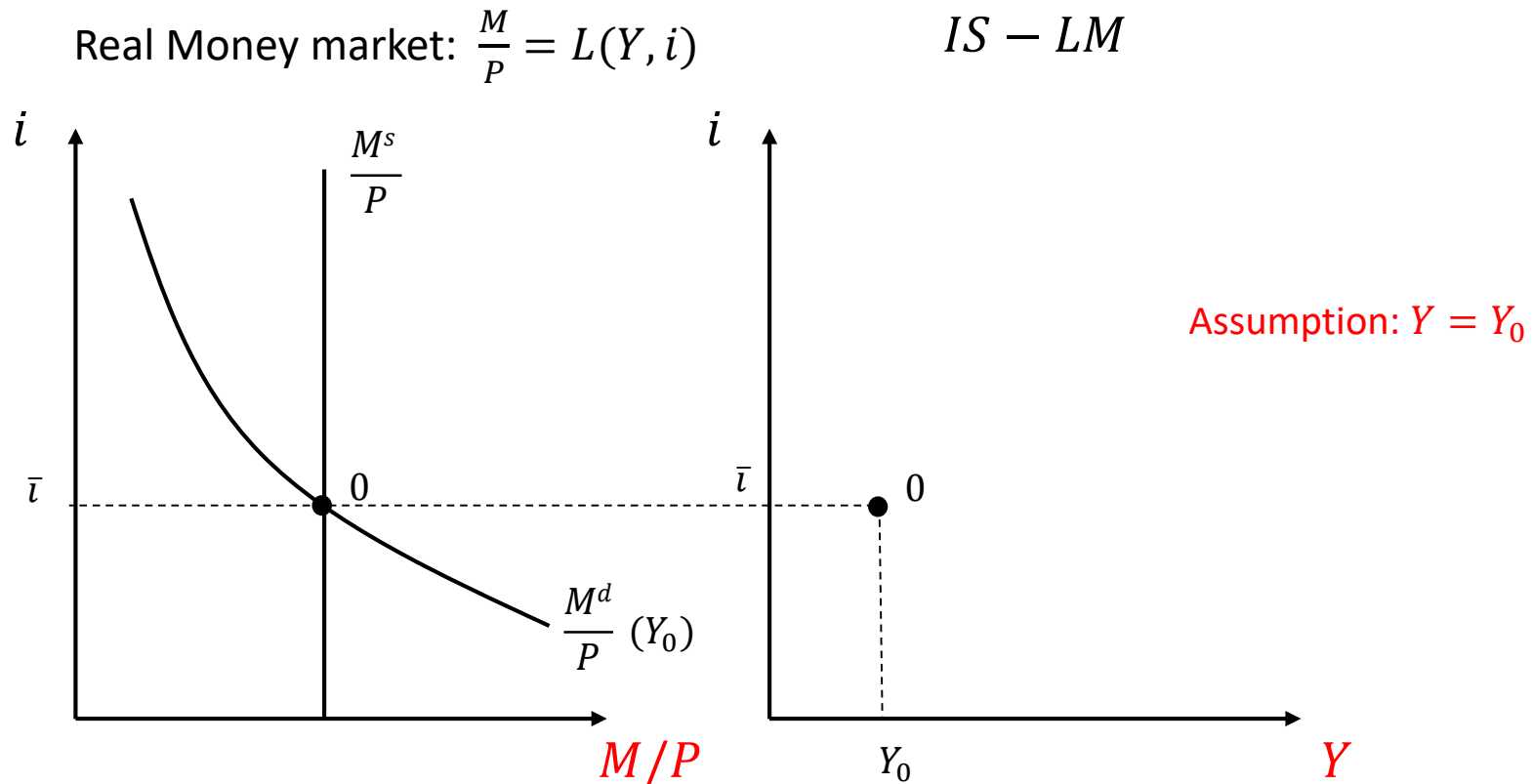
LM – STANDARD CASE

BC sets i and adjusts $M^s \Rightarrow i$ is exogenously chosen and money supply is endogenous wrt the money demand

What shape will the LM curve have?

LM STANDARD CASE (i exogenous): graphical construction

BC sets $i = \bar{i}$



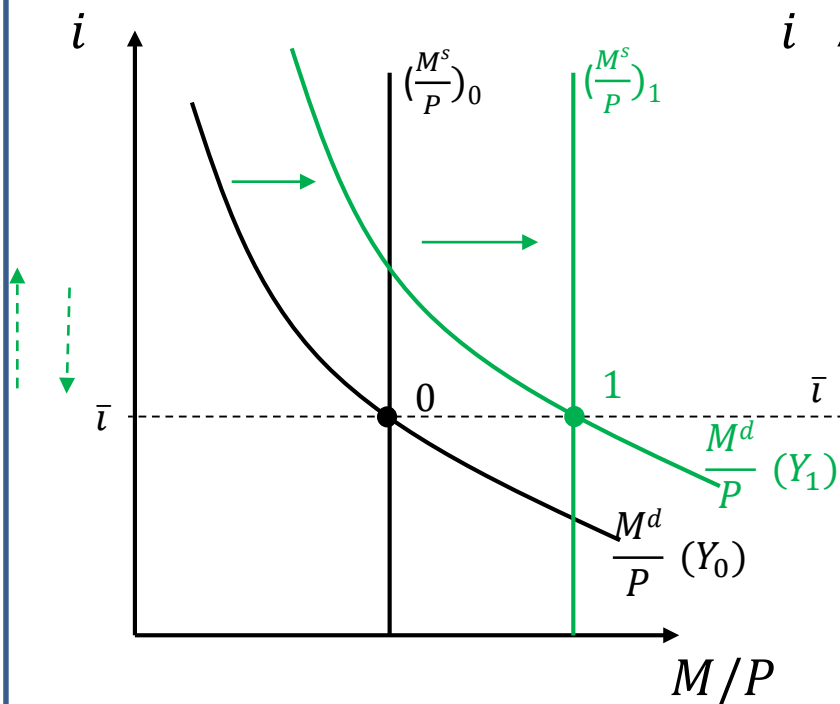
LM: STANDARD CASE

What happens on the money market when Y changes and the CB has an interest rate target?

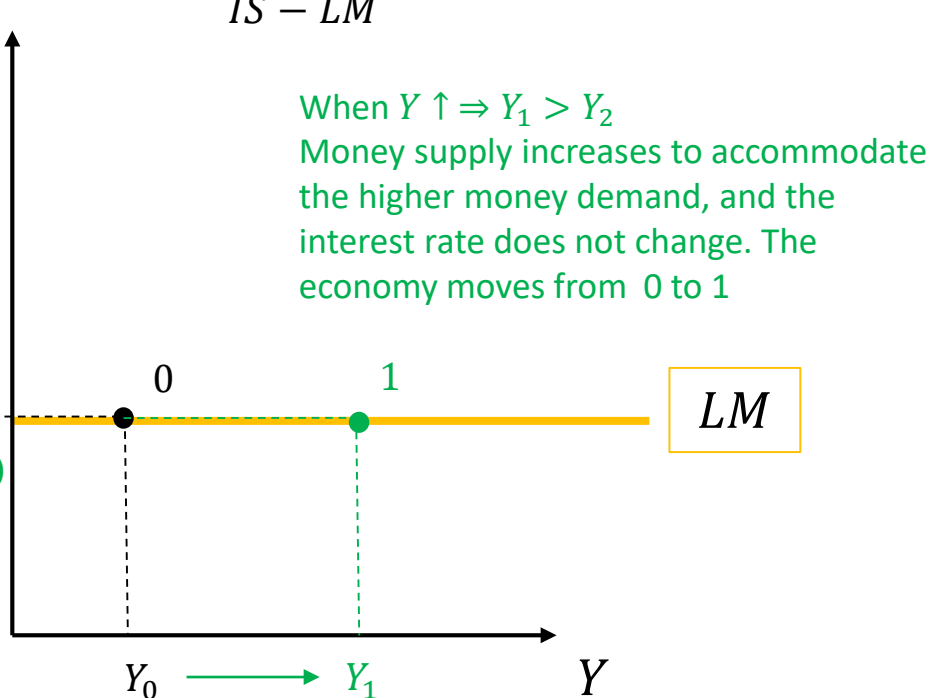
$Y \uparrow \Rightarrow M^d \uparrow \Rightarrow$ upward pressures on i in the money market

To keep i constant at $\bar{i}$, the CB has to $\uparrow M^S$

Money market



IS – LM

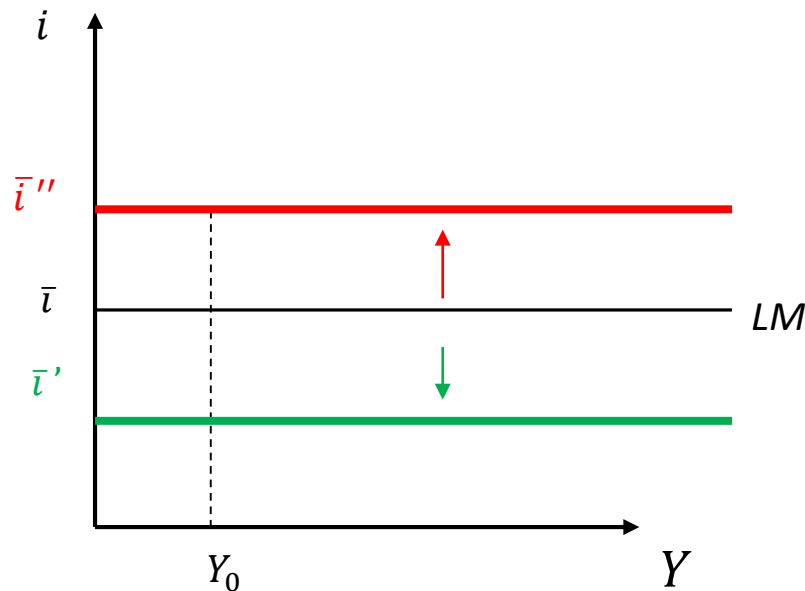


When $Y \uparrow \Rightarrow Y_1 > Y_2$
Money supply increases to accommodate
the higher money demand, and the
interest rate does not change. The
economy moves from 0 to 1

$$LM: i = \bar{i}$$

MONETARY POLICY EFFECT ON LM

Changing the liquidity level in the economy changes the interest rate that ensures equilibrium in the money market $\Rightarrow$ the LM curve shifts



Contractionary MP: CB reduces liquidity by selling bonds $\Rightarrow \epsilon P_t \downarrow \Rightarrow i \uparrow$

Increase in the policy target interest rate $\Rightarrow$ the LM shifts up

Expansionary MP: CB injects liquidity by buying bonds $\Rightarrow \epsilon P_t \uparrow \Rightarrow i \downarrow$
Decrease in the policy target interest rate $\Rightarrow$ the LM shifts down

If the CB has a target in terms of monetary stock, the LM curve

- a) is downward sloping.
- b) is upward sloping.**
- c) does not depend on the interest rate.
- d) has a slope lower than 1

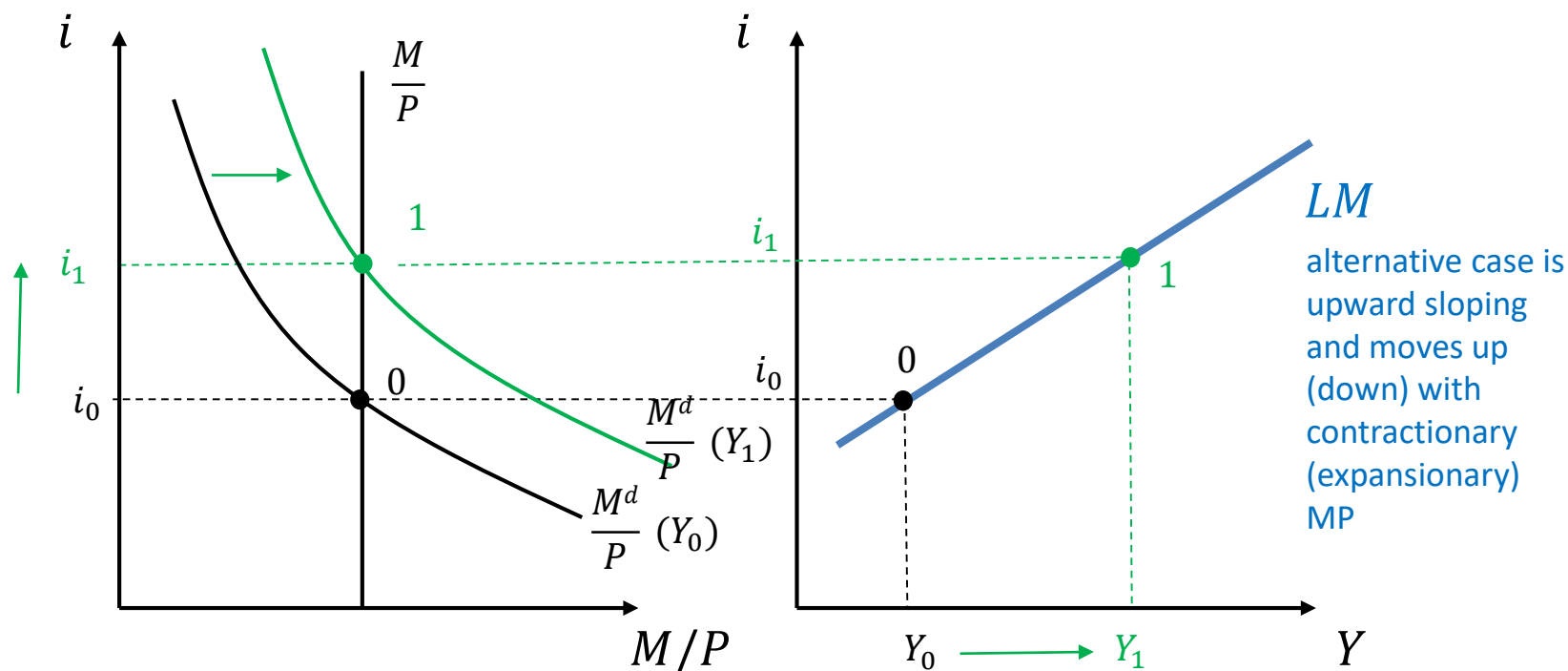
What happens on the money market when income changes and the CB sets M?

$Y \uparrow \Rightarrow M^D \uparrow \Rightarrow$ excess money demand $\Rightarrow$ agents sell bonds, their price decreases and the interest rate increases since the CB does not intervene in the money market and simply keeps $\Rightarrow i \uparrow$

Assumption: $Y \uparrow \Rightarrow Y_1 > Y_2$

Real Money market

IS – LM



THE IS CURVE

EQUILIBRIUM IN THE GOODS MARKET

Aggregate supply must be equal to aggregate demand of goods and services or investment (I) must be equal to national saving (S)

ASSUMPTIONS

- 1 good
- fixed P
- Closed economy
- $C = C(Y - T)$
- G & T : exogenous
- I endogenous : $I = I(Y, i)$ Remember I is firms' investment in physical capital in new machines and plants

+ -

←
If $Y \uparrow$, more income in the economy and more demand, firms sale more, higher incentive to invest in new machines and plants $\Rightarrow \uparrow I$

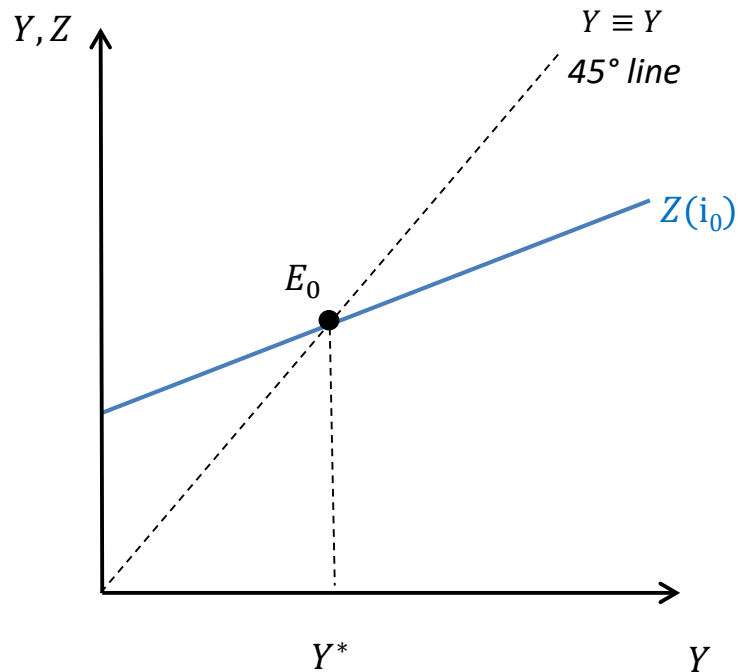
→
If $i \uparrow$, less attractive to borrow money to buy new machines $\Rightarrow$ firms without own capital tend to invest less $\Rightarrow \downarrow I$

i is the **cost of borrowing** :
it affects private decisions about spending

Underlying assumptions:
- Same i we considered for bonds
- No borrowing constraints

$$\text{Aggregate demand } Z(i) = C(Y - T) + I(Y, i) + G$$

The goods market is in equilibrium, when, at the prevailing interest rate (i_0), aggregate supply is equal to aggregate demand

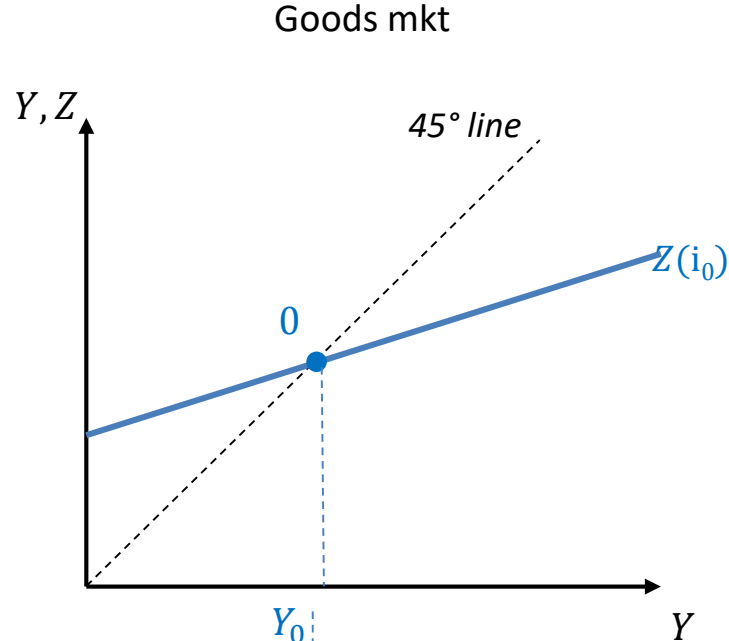


$$Y = Z(i_0)$$

ZZ properties:

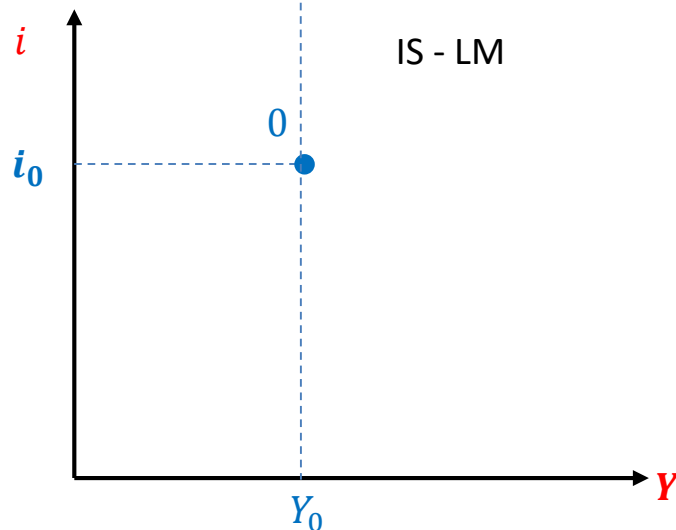
- 1) Positive slope but lower than 1: If $Y \uparrow \Rightarrow \uparrow C$ and $\uparrow I$, but part of the additional income is saved
- 2) Position depends on exogenous variables: The same as for the goods market but also the interest rate

IS CURVE: graphical construction



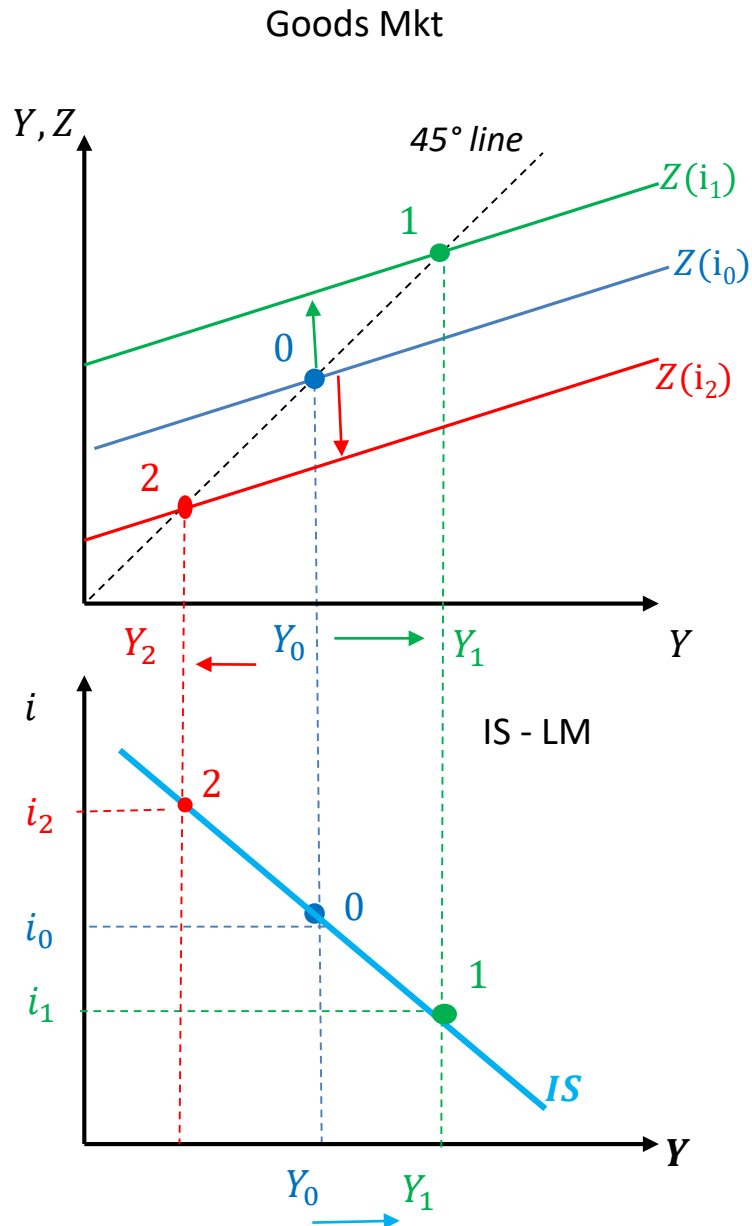
Given the interest rate i_0 , equilibrium income on the goods market is Y_0

Combination 0 belongs to the IS curve



*What happens to the goods market equilibrium if the interest rate changes?
The equilibrium changes*

The IS curve is constructed by finding the goods market equilibrium when i changes



Assume $i \downarrow: i_1 < i_0$

If $i \downarrow \Rightarrow I \uparrow \Rightarrow Z \uparrow \Rightarrow$ in the goods market there is an excess demand, and in order to restore the equilibrium, firms produce more. The economy moves from 0 to 1.

Assume $i \uparrow: i_2 > i_0$

If i increases, firms invest less since it is more costly. The aggregate demand decreases and in the goods market there is an excess supply: firms will produce less. The economy moves from 0 to 2.

...repeating for any possible value of i we find infinite points that belong to the **IS CURVE**

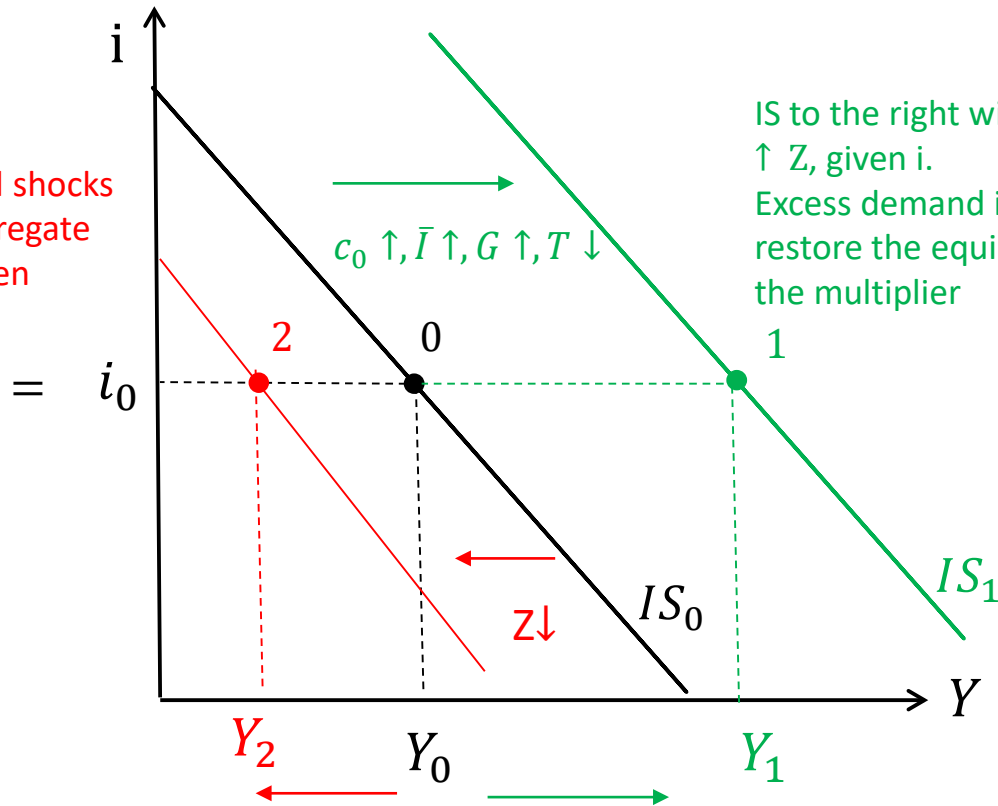
All the pairs (Y, i) where the goods market is in equilibrium

If i changes, we move **ALONG** the IS

IS POSITION

depends on exogenous variables influencing the output level in the goods market, given i

IS to the left with negative demand shocks reducing the aggregate demand, at a given interest rate



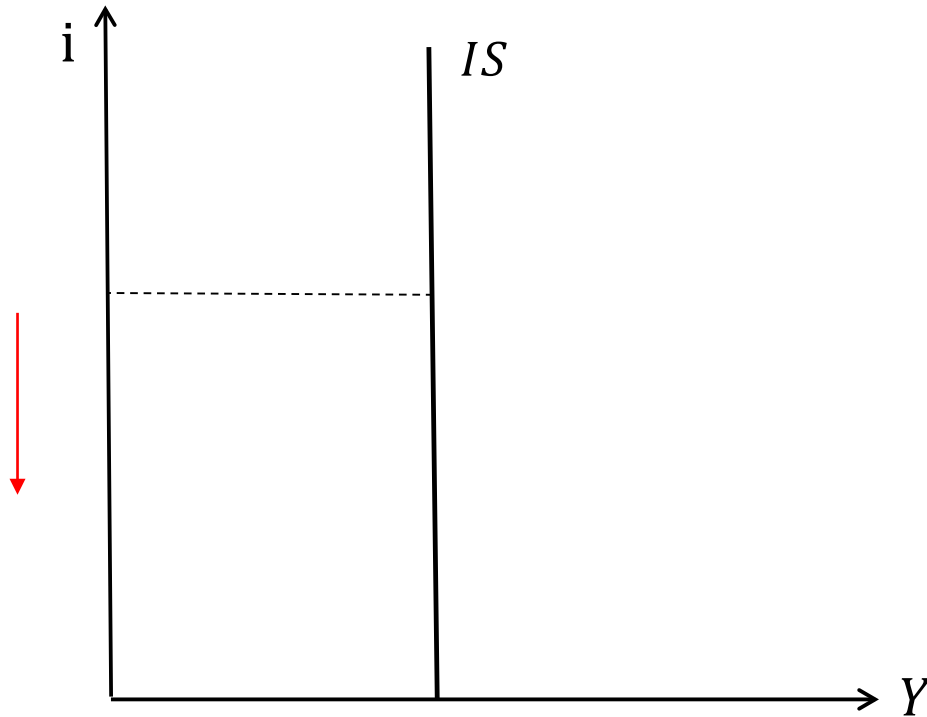
IS to the right with positive demand shocks that $\uparrow Z$, given i .
Excess demand in the goods market $\Rightarrow$ to restore the equilibrium Y must increase, through the multiplier

TRUE OR FALSE:

In a country, the IS curve will be vertical if all components of demand depend negatively on the interest rate.

False!

IS VERTICAL



No effect of i on Y :

if $i \downarrow \Rightarrow Z$ does not change since no component of the demanda is affected $\Rightarrow Y$ stays constant

=

IS rigid if
No component of Z depends on i

TRUE OR FALSE:

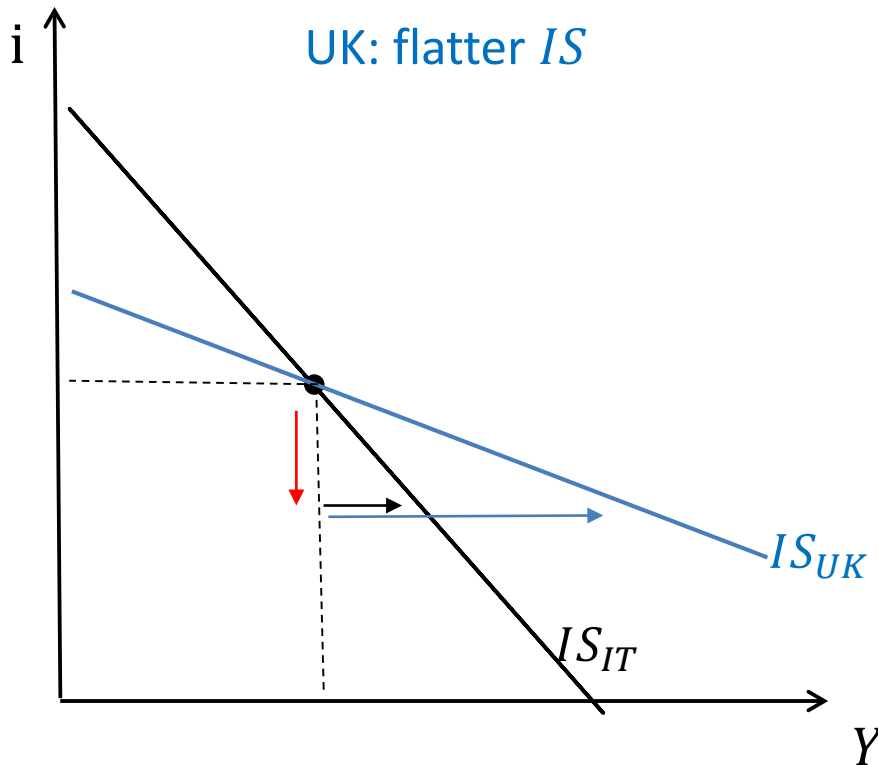
The UK will have a flatter IS curve than Italy **only if** more components of demand depend negatively on the interest rate.

False!

IS FLATTER

Italy: steeper IS

UK: flatter IS



Given the same reduction in i , in UK Z varies more
⇒ the equilibrium level of Y varies more
⇒ Y reacts more when i changes in UK than in IT

- 1) because more components of the demand change when i changes
- 2) because I changes more when i changes
- 3) because the multiplier effect of the goods market is stronger and after the initial change in the demand the indirect effect is stronger

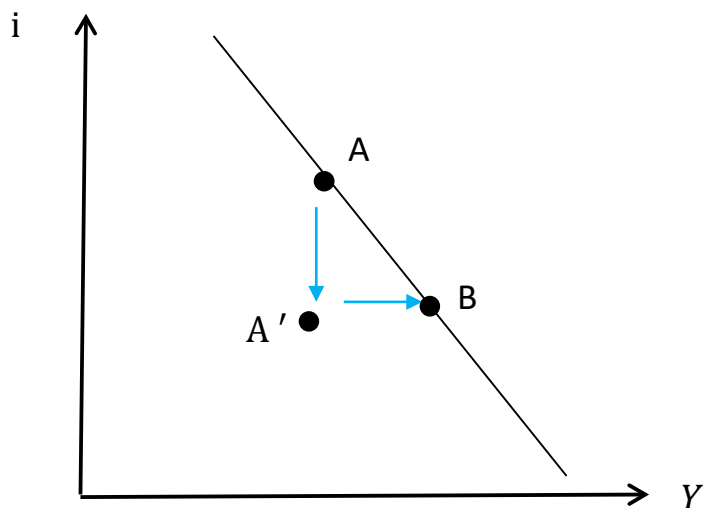
IS flatter if

- I more sensitive to i
- more components of Z depend on i
- I and C more sensitive to Y

IS SLOPE

- Captures the relationship btw i and Y
- Basic case: negative slope since there is a trade off btw i and Y through some components of the demand (in the basic case only investment)

– If $i \downarrow \Rightarrow I \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow$



In A and B the goods market is in equilibrium

When I decreases, the economy moves in A' where there is an excess demand and the market is not in equilibrium $\Rightarrow$ in order to restore the equilibrium Y has to increase up to point B

...but the slope will depend on the assumptions we make!

SUMMING UP

- *LM*: interest rate that keeps financial markets in equilibrium for any value of Y
 - Derivation
 - Shifts
 - Standard case vs. alternative case
- *IS*: income that keeps goods market in equilibrium for any value of the interest rate
 - Endogenous investment
 - Derivation
 - Shifts
 - Slope

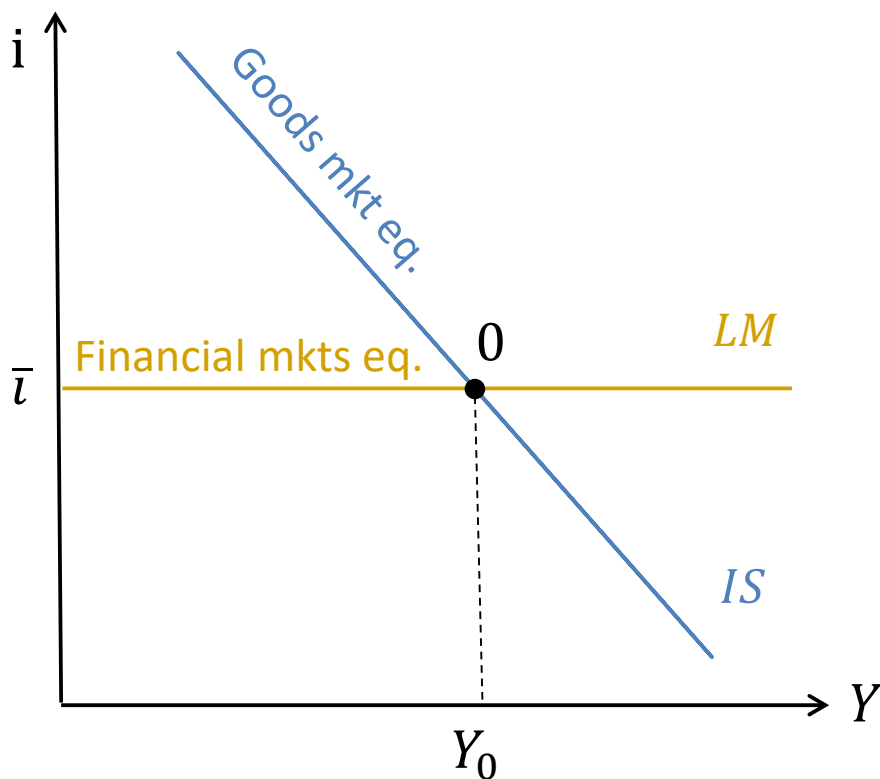
SHORT RUN MACROECONOMIC EQUILIBRIUM:

All markets are in equilibrium

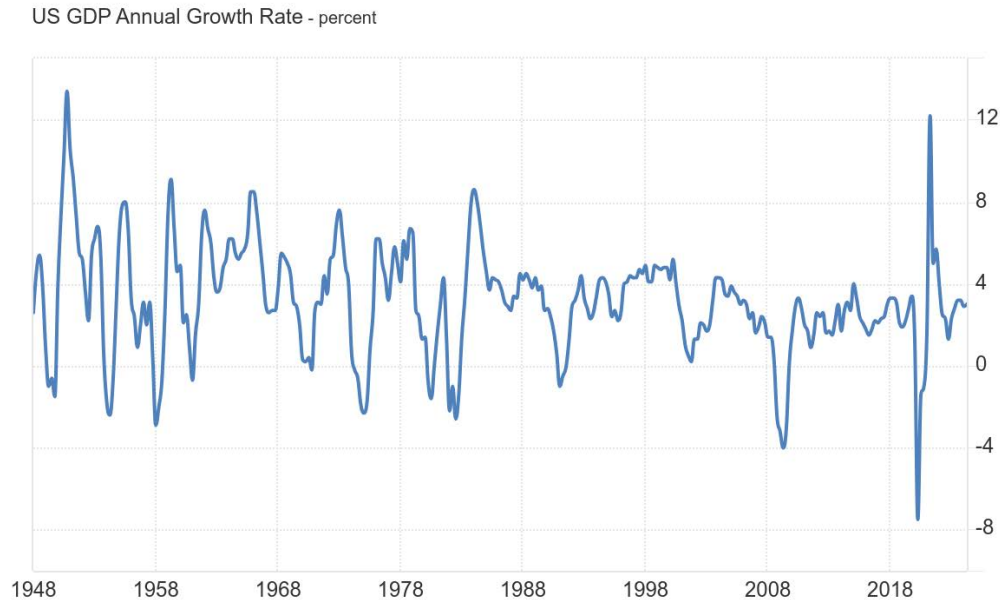
$IS - LM$: STANDARD CASE

$$LM: i = \bar{i}$$

$$IS: Y = C(Y - T) + I(Y, i) + G$$



SHORT RUN ECONOMIC FLUCTUATIONS



Source: tradingeconomics.com | U.S. Bureau of Economic Analysis

- In the short run, many shocks hit real and/or financial markets
 - The **macro equilibrium changes**
 - We have **economic fluctuations**
 - Policy makers look for **interventions to stabilize** the economy and avoid negative effects

A FISCAL POLICY SHOCK

Expansionary: $G \uparrow$; $T \downarrow$

Contractionary: $G \downarrow$; $T \uparrow$

Change in the behavior of the public sector.

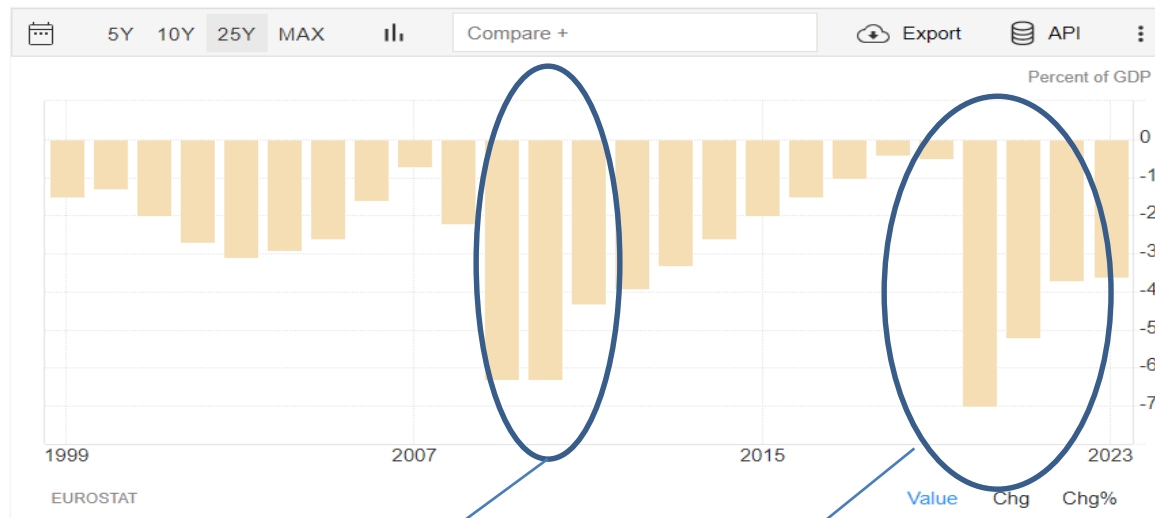
What happens to the macro equilibrium?

What are the effects on the demand and on saving?

Euro Area Government Budget

Summary Forecast Calendar Alerts Download ▾

Euro Area recorded a Government Budget deficit equal to 3.60 percent of the country's Gross Domestic Product in 2023. Government Budget in Euro Area averaged -3.08 percent of GDP from 1995 until 2023, reaching an all time high of -0.40 percent of GDP in 2018 and a record low of -7.30 percent of GDP in 1995. source: EUROSTAT

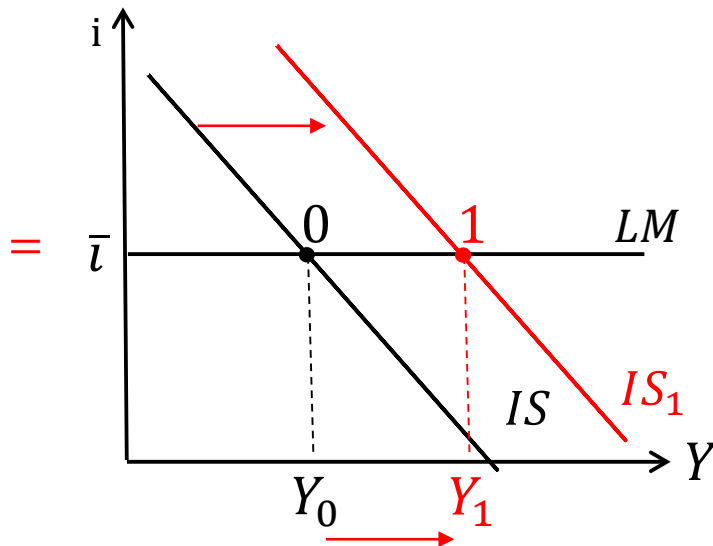


During the 2008 Crisis and the Covid crisis, public deficit increased a lot (Maastricht parameters (Deficit/GDP<3%)) since government raised public expenditure.

Why during recessions expansionary FP are implemented?

Use the IS-LM model to explain what we could expect to be the effect on the macro equilibrium?

Assume the **government** increases public expenditure: $G \uparrow$



Graphically:

- IS to the right: positive demand shock $G \uparrow \Rightarrow Z \uparrow \Rightarrow$ at the same interest rate $Y \uparrow$

- LM =

Equilibrium moves from 0 to 1

$Y \uparrow \quad \bar{i} =$

Economic intuition

Effect on the real market:

- $G \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow \Rightarrow C \text{ \& } I \uparrow \Rightarrow Z \uparrow \Rightarrow \dots \Rightarrow Y \uparrow \uparrow$:

the increase in G raises demand and production increases more than proportionally thanks to the goods market multiplier. Indeed, as income increases, consumption & investment go up too, generating a further increase in the demand and in income

$$\Delta Y > \Delta G$$

Effect on the financial market:

$$Y \uparrow \Rightarrow M^D \uparrow \Rightarrow M^S \uparrow \Rightarrow i = :$$

more money needed for transactions $\Rightarrow$ upward pressure on i since individuals sell bonds $\Rightarrow \epsilon P_B \downarrow \Rightarrow i \uparrow$

$\Rightarrow$ the CB to keep the interest rate constant increases the monetary base to equate the money supply to the

higher money demand $\Rightarrow$ CB buys bonds $\epsilon P_B \uparrow \Rightarrow i \downarrow$

$\Rightarrow$ the interest rate does not change

AGGREGATE DEMAND

How does the composition of the aggregate demand change in the new equilibrium?

At point 1 compared to point 0

- The level of the demand will be higher since income is higher ($Y=Z$ and since $Y \uparrow$, also $Z \uparrow$)
- The composition of the demand will be different:
 - Consumption **Increases as $Y \uparrow$ and T does not change**
 - Investment **Increases as $Y \uparrow$ and i does not change**
 - Government expenditure **Increases by assumption**

NATIONAL SAVING

What happens to national saving?

- Private saving, S :
 - Increases as $Y \uparrow$ and T does not change
- Public saving, $S^G = T - G$:
 - Decreases, as $G \uparrow$ and deficit increases

What is the net impact of the two changes?

We know that $I \uparrow$ and $I = S + S^G \Rightarrow$

$$|\Delta S| > |S^G|$$

The increase in private saving more than compensate the reduction in public saving
 $\Rightarrow$ national saving increases

REAL MONEY DEMAND

$$\frac{M}{P} = L(Y, i)$$

What happens to real money demand?

- Increases as $Y \uparrow$ and i does not change
- It increases by the same amount of the money supply

$$\Delta H > 0 \quad \Delta \frac{M^s}{P} = m^M \Delta H$$

FISCAL POLICY CONCLUSION

- A fiscal policy or any shock affecting the equilibrium on the **goods market** shift the **IS** (left or right!)
- To keep the interest rate fixed, an open market operation by the Central Bank is needed
 - In the alternative case the CB does not intervene
- An expansionary fiscal policy boosts GDP => it is typically implemented during crisis
- A contractionary fiscal policy reduces GDP => can be implemented when the economy is growing sufficiently, and the public sector wants to reduce public deficit

MAGNITUDE OF THE IMPACT

Fiscal policy multiplier: by how much Y changes if autonomous spending changes by a given amount

- Larger impact of ΔA if the multiplier of the goods market is higher
- If C and I are more sensitive to Y : indirect effect of the demand on equilibrium income for the goods market is higher

A MONETARY POLICY SHOCK

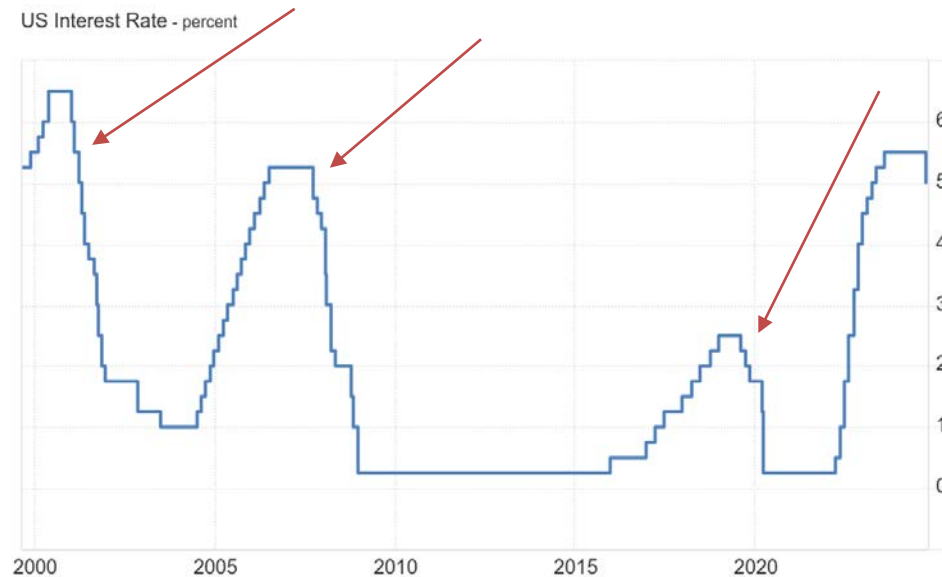
Expansionary: $i \downarrow$

Contractionary: $i \uparrow$

Change in the behavior of the CB.

What happens to the macro equilibrium?

What are the effects on the demand and on saving?



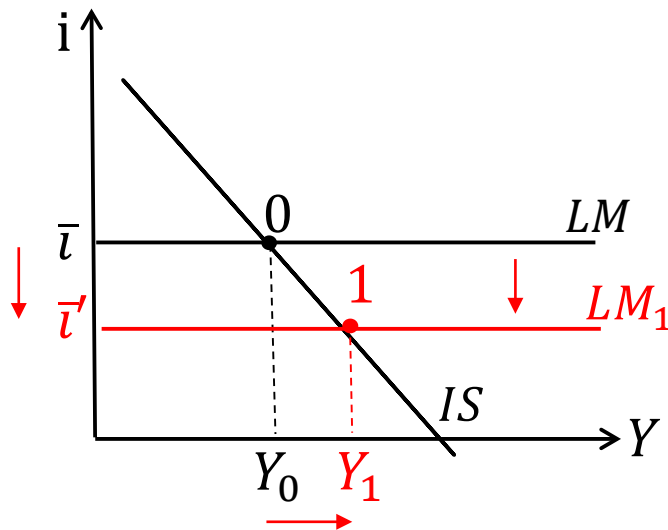
Source: tradingeconomics.com | Federal Reserve

During the crisis, CB cuts the interest rate.

Why during recessions expansionary MP are implemented?

Use the IS-LM model to explain what we could expect to be the effect on the macro equilibrium?

Assume the **Central Bank** implements an expansionary monetary policy: $\bar{i} \downarrow$



Graphically:

- $IS =$
- **LM DOWN:** CB buys bonds on the open market and pays with new liquidity. Bond's price increases and interest rate decreases

Equilibrium moves from 0 to 1

$$Y \uparrow \quad \bar{i} =$$

Economic intuition

- **Effect on the financial market:**

- $H \uparrow \Rightarrow \text{€}P_B \uparrow \Rightarrow i \downarrow$:

Expansionary OMO

$\Rightarrow$ CB buys and since the demand for bonds increases in the secondary market their price increases

$\Rightarrow$ given the bonds face value, their return decreases

- **Effect on the real market:**

- $i \downarrow \Rightarrow I \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow \Rightarrow C \& I \uparrow \Rightarrow Z \uparrow \Rightarrow \dots \Rightarrow Y \uparrow \uparrow$:

The interest rate decline reduces the cost of borrowing and firms invest more. This increases the demand and then production.

The increase in production is more than proportionally thanks to the goods market multiplier. Indeed, as income increases, consumption & investment go up too, generating a further increase in the demand and in income

AGGREGATE DEMAND

How does the composition of the aggregate demand change in the new equilibrium?

At point 1 compared to point 0:

- The level of the demand will be higher since income is higher ($Y=Z$ and since $Y \uparrow$, also $Z \uparrow$)
- The composition of the demand will be different:
 - Consumption: Increases as $Y \uparrow$ and T does not change
 - Investment: Increases as $Y \uparrow$ & $i \downarrow$
 - Government expenditure Constant by assumption

NATIONAL SAVING

What happens to national saving?

- Private saving, S :
 - Increases as $Y \uparrow$ and T does not change
- Public saving, $S^G = T - G$:
 - Constant as well as public deficit

What is the net impact of the two changes?

We know that $I \uparrow$ and $I = S + S^G \Rightarrow$

$$|\Delta S| = |\Delta I|$$

The increase in private saving is exactly equal to the increase in investment
 $\Rightarrow$ national saving increases

REAL MONEY DEMAND

$$\frac{M}{P} = L(Y, i)$$

What happens to real money demand?

- Increases as $Y \uparrow$ and i decreases (more need of liquidity for transactions and bonds are less attractive for investors)
- It increases by the same amount of the money supply for the OMO

$$\Delta H > 0 \quad \Delta \frac{M^S}{P} = m^M \Delta H$$

Note that $M^S \uparrow$ more than the amount required to decrease i given Y
In fact, as $Y \uparrow \Rightarrow M^D \uparrow$

MONETARY POLICY: CONCLUSION

- A **monetary policy** shifts the ***LM*** (up or down!)
- An expansionary monetary policy reduces the interest rate and boosts GDP
- A contractionary monetary policy increases the interest rate and reduces GDP

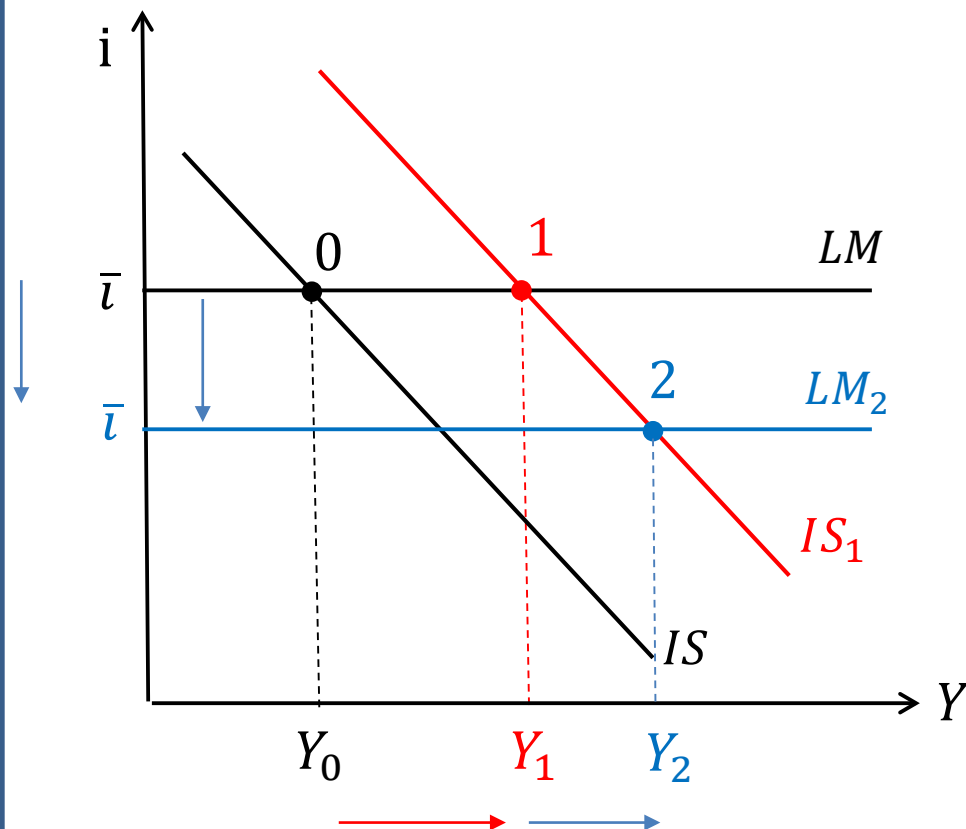
A POLICY MIX: USING FP AND MP TOGETHER TO REACH A SPECIFIC GOAL

Imagine that our economy is facing a very **severe recession**

The economy is in equilibrium at a low level of GDP, Y_0 . The government and the CB can help the economy to recover by implementing a **coordinated policy intervention** to boost the economy:

Policy Mix:

- An expansionary FP: $G \uparrow$
- An expansionary MP: $\bar{\tau} \downarrow$



Government: $\uparrow G$

The expansionary fiscal policy shifts the *IS* to the RIGHT

Equilibrium moves from 0 to 1

$$Y \uparrow \quad \bar{i} =$$

CB: $\downarrow i$

The expansionary monetary shifts the *LM* DOWN

Equilibrium moves from 1 to 2

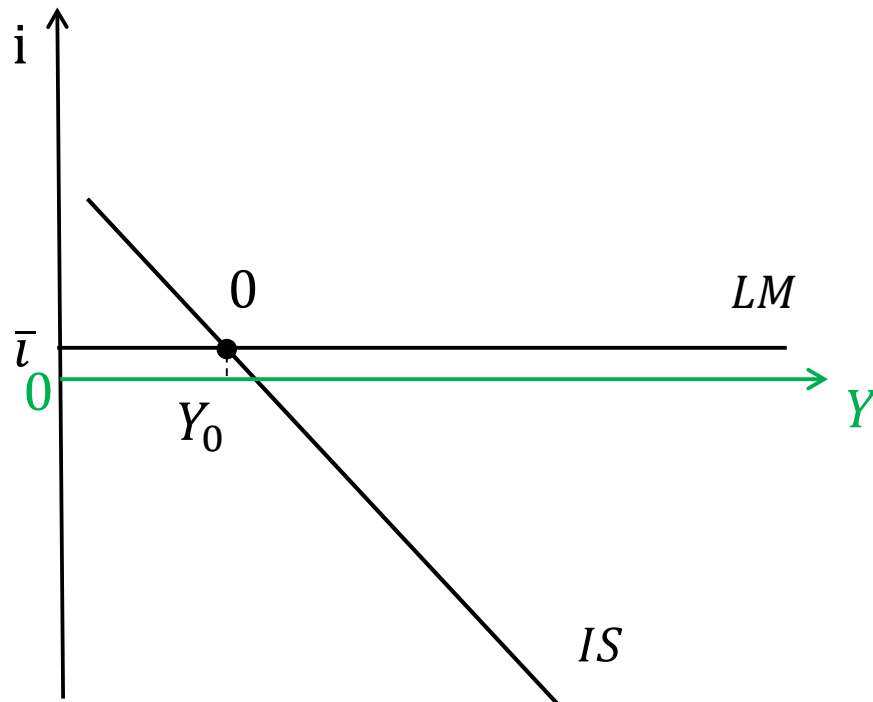
$$Y \uparrow \uparrow \quad \bar{i} \downarrow$$

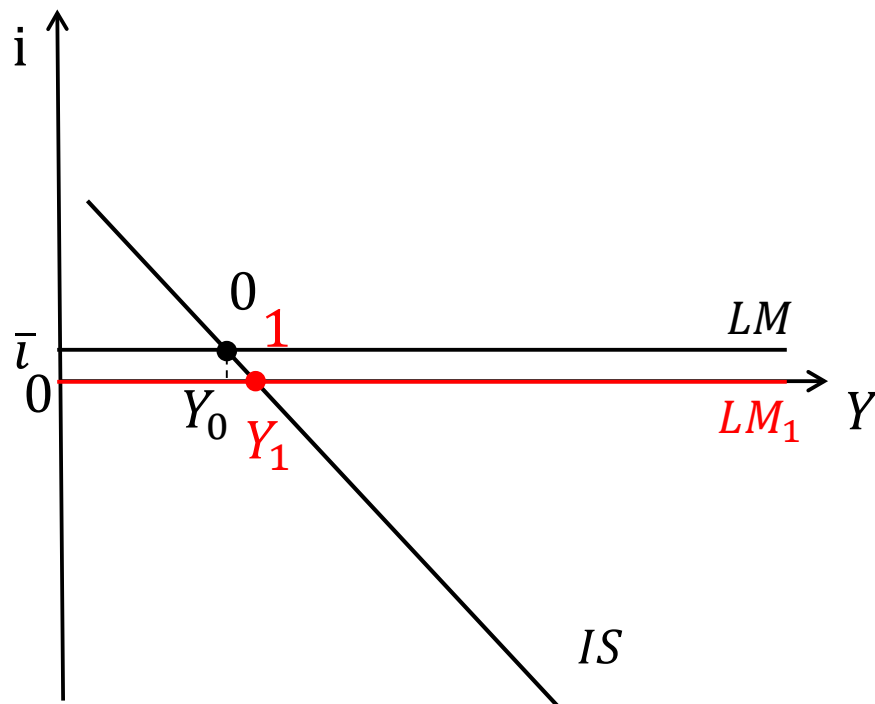
The combined use of fiscal and monetary policy can push GDP at a higher level

POLICIES AT THE ZERO-LOWER BOUND

POLICIES AT THE ZERO – LOWER BOUND

Assume that in the initial equilibrium 0 the interest rate is already close to zero

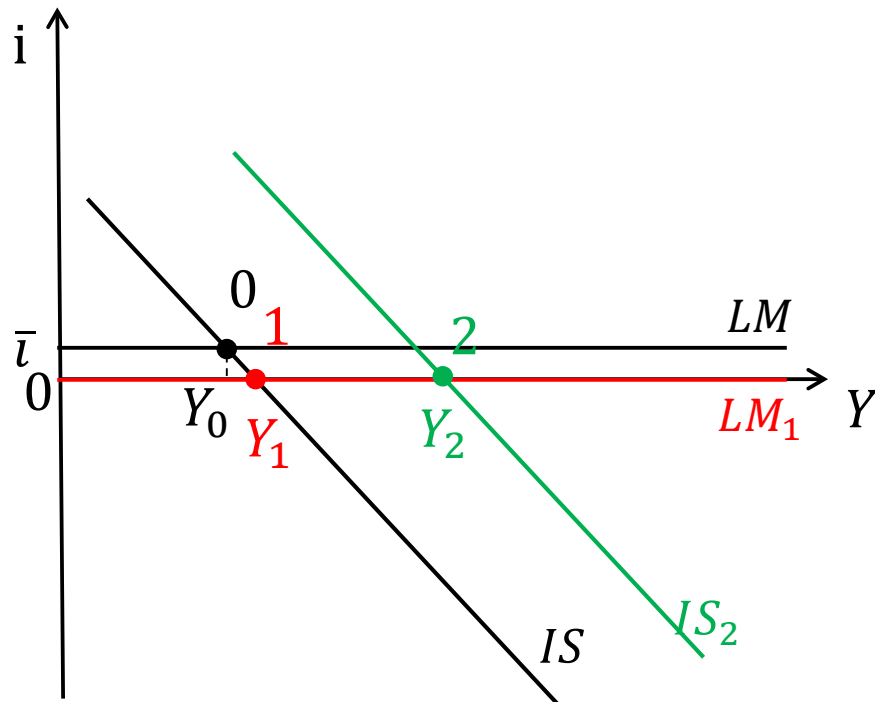




Small space to implement MP. At most i can be zero.

By reducing $\bar{i}$ to zero, the increase in GDP is limited

Unconventional monetary policies?



After point 1 only one policy tool, **fiscal policy**, can then be used to increase GDP!

- ... but in some case is not possible:
- if the Government is already running a huge public **deficit**
 - If the country has accumulated a large public **debt**

ECONOMIC POLICY MIX

WOW! It seems *“a kind of magic”* !!!



By using the policies, we can “choose” the value of GDP...

Then, why did economies suffer prolonged crises?

Both FP and MP face constraints

POLICY CONSTRAINTS

Fiscal policy

- Uncertainty of the impact: It depends on the parameters of the economy/agents' behavior
- It takes time to decide a policy at the Government level
- Problems in financing the policy
 - Public balance vs. debt accumulation
 - Debt sustainability

Monetary policy

- Transmission mechanism (lag in real impact)
 - How long does it take for banks to change the interest rate paid by firms?
- Zero-lower bound
 - $i \geq 0$
 - What if i is already very low?

QUICK CHECK

The CB of a common monetary area purchases bonds for €1 million.

In country Alpha, consumption depends positively on disposable income, and investment depends positively on income and negatively on the interest rate.

In country Beta, consumption depends positively on disposable income and negatively on the interest rate, while investment depends positively on income and negatively on the interest rate.

Hence:

- a) GDP will increase more in Alpha
- b) GDP will increase less in Alpha**
- c) GDP will not change in Alpha
- d) GDP will not change in Beta

QUICK CHECK

If an economy is in a liquidity trap,

a) only a restrictive monetary policy, which drains the excess liquidity existing in the economic system, could increase equilibrium income.

b) the increase in public purchases of goods and services increases the equilibrium income.

c) only an expansionary monetary policy, capable of satisfying the large demand for liquid assets existing in the economic system, could increase equilibrium income.

d) the increase in public purchases of goods and services decreases equilibrium income.

QUICK CHECK

Assume G and T increase by the same amount. The short run equilibrium for output will _____ if the central bank has a policy rate target than if the central bank chooses the money supply.

- a) increase less
- b) decrease more
- c) decrease less
- d) increase more**

TAKE AWAY AND KEY CONCEPTS

- IS Curve
 - Meaning
 - Slope
 - Position
- LM Curve
 - Meaning
 - Slope
 - Position
 - Standard case and alternative case
- Short-run equilibrium
 - Fiscal Policy:
 - Expansionary: Increases income and production by shifting the IS curve to the right.
 - Contractionary: Reduces income and production by shifting the IS curve to the left.
 - Monetary Policy:
 - Expansionary: Increases income by lowering interest rates and shifting the LM curve downward.
 - Contractionary: Reduces income by increasing interest rates and shifting the LM curve upward.
 - Policy Mix
- Policies at the ZLB
- Constraints and Limits